

Chemistry Paper 1

Solved

Part I and every question of Part II, 70 marks

Answer · Why · NCERT page

Formula shown for every numerical

All choices worked

How each answer is laid out. The question as set, then — for a numerical — the **formula used** in a box of its own before any arithmetic, then the **answer**, then a short **why**, then the **NCERT reference** as *Ch N, p. NN*.

This paper is on the old syllabus, and that matters. It was set in 2016. The 2023 rationalisation removed **Solid State**, **Ionic Equilibrium** (buffers, solubility product), **Surface Chemistry**, the **p-Block Elements**, **metallurgy** and **Polymers** from the Class XII book altogether. Questions on those topics therefore have **no NCERT page**, and the reference line says so rather than guessing. Roughly a third of this paper is in that position — worth knowing before you use it as revision for the current course.

Page numbers are the pages printed in the NCERT Class XII textbook (Parts I and II, 2026–27 reprint):

Ch 1 pp. 1–30, **Ch 2** 31–60, **Ch 3** 61–88, **Ch 4** 89–117, **Ch 5** 118–140, **Ch 6** 159–192, **Ch 7** 193–226, **Ch 8** 227–258, **Ch 9** 259–280, **Ch 10** 281–302.

The rubric. Answer all of Part I and six from Part II — two from each of Sections A, B and C. Every question is worked here, so you can choose freely.

Part I · 20 marks

Question 1, compulsory: five fill-in-the-blanks, five to match, five to explain and five multiple choice.

Question 1 (a) · Fill in the blanks

5

1(a)(i) Ionic equilibrium

[1]

Solubility of calcium oxalate is _____ in the presence of ammonium oxalate because of _____.

Answer · decreased; because of the **common ion effect**.

Why · Ammonium oxalate supplies extra $\text{C}_2\text{O}_4^{2-}$, an ion the calcium oxalate equilibrium already contains. By Le Chatelier's principle the equilibrium $\text{CaC}_2\text{O}_4(\text{s}) \rightleftharpoons \text{Ca}^{2+} + \text{C}_2\text{O}_4^{2-}$ shifts to the left, so less dissolves — the ionic product must stay equal to K_{sp} .

NCERT · — (not in the NCERT Class XII book) *Ionic equilibrium is a Class XI topic and is not in the Class XII book.*

1(a)(ii) Alcohols and Phenols

[1]

Ethanol is _____ to litmus while phenol is _____ to litmus.

Answer · neutral; acidic.

Why · Phenol's conjugate base, the **phenoxide** ion, is stabilised by delocalisation of the negative charge into the benzene ring, so phenol gives up its proton ($\text{p}K_{\text{a}} \approx 10$). An ethoxide ion has no such delocalisation — the alkyl group is even +I and pushes charge onto the oxygen — so ethanol ($\text{p}K_{\text{a}} \approx 16$) is effectively neutral.

NCERT · Ch 7, p. 206

1(a)(iii) Solutions

[1]

The _____ pressure of an aqueous solution of 0.1 M cane sugar is _____ than that of pure water measured at the same temperature.

Answer · vapour pressure; lesser than.

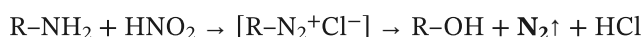
Why · Raoult's law: the vapour pressure of the solvent is proportional to its mole fraction. Dissolving a non-volatile solute lowers that mole fraction and, on the surface picture, sugar molecules occupy part of the surface so fewer water molecules can escape. Hence the **lowering of vapour pressure**, the colligative property from which the other three follow.

NCERT · Ch 1, p. 15

1(a)(iv) Amines

[1]

On treating a _____ with _____ nitrogen gas is evolved.



Answer · a **primary amine** with **nitrous acid**.

Why · An aliphatic primary amine gives a diazonium salt that is far too unstable to survive, so it decomposes at once, releasing N_2 quantitatively and leaving the alcohol. Secondary amines give a nitrosamine and tertiary amines a salt — neither evolves nitrogen, which is what makes this a test.

NCERT · Ch 9, p. 274

1(a)(v) Electrochemistry

[1]

The more _____ the reduction potential of a metal, the _____ is its ability to displace hydrogen from acids.

Answer · the more **negative**; the **greater**.

Why · Displacing hydrogen means the metal is oxidised while H^+ is reduced. For that to be spontaneous the metal must be the *better reducing agent* of the two, i.e. sit below hydrogen in the series — which is exactly what a more negative E°_{red} means. Zinc (-0.76 V) displaces hydrogen; copper ($+0.34\text{ V}$) does not.

NCERT · Ch 2, p. 35

Question 1 (b) · Match the following

5

1(b) Mixed

[5]

Match column A with column B: **i)** Nernst equation **ii)** Ostwald's dilution law **iii)** Lewis base **iv)** Zwitter ion **v)** Hinsberg's reagent — **a)** Glycine **b)** Ammonia **c)** Amine **d)** Electrode potential **e)** Weak electrolyte

Answer ·

	Column A	→	Column B
i	Nernst equation	d	Electrode potential
ii	Ostwald's dilution law	e	Weak electrolyte
iii	Lewis base	b	Ammonia
iv	Zwitter ion	a	Glycine
v	Hinsberg's reagent	c	Amine

Why · **i-d** the Nernst equation gives the electrode potential at any concentration. **ii-e** Ostwald's dilution law relates the degree of dissociation of a *weak* electrolyte to its dilution. **iii-b** a Lewis base donates a lone pair, and NH_3 is the standard example. **iv-a** glycine, like every α -amino acid, exists as the internally neutralised zwitter ion. **v-c** Hinsberg's reagent (benzenesulphonyl chloride) distinguishes 1° , 2° and 3° amines.

NCERT · Ch 2, p. 38 — for the Nernst equation; Hinsberg's reagent is at Ch 9, p. 271 and the zwitter ion at Ch 10, p. 292

Question 1 (c) · Explain the following facts

5

1(c)(i) Phenols

[1]

Unlike most phenolic compounds, 2,4,6-trinitrophenol is soluble in sodium bicarbonate solution.

Answer · 2,4,6-trinitrophenol (**picric acid**) is a genuinely strong acid, $pK_a \approx 0.4$ — comparable with a mineral acid. It is therefore strong enough to liberate CO_2 from $NaHCO_3$ and dissolve as its sodium salt. An ordinary phenol ($pK_a \approx 10$) is far too weak to do so and dissolves only in the stronger base $NaOH$.

Why · The three $-NO_2$ groups are each $-I$ and $-R$, and two of them are *ortho* and one *para*, so all three can delocalise the phenoxide's negative charge by resonance. That is an enormous stabilisation of the anion, and a well-stabilised anion means a strong acid.

NCERT · Ch 7, p. 206

This is the standard way of separating a carboxylic acid from a phenol — $NaHCO_3$ dissolves the acid only. Picric acid is the exception that proves the rule.

1(c)(ii) Electrochemistry

[1]

Specific conductance of a solution decreases with dilution.

Answer · Specific conductance (conductivity, κ) is the conductance of **one unit volume** of solution — a 1 cm cube. On dilution the *number of ions in that fixed volume* falls, so κ falls.

Why · The contrast with molar conductivity is the point. $\Lambda_m = \kappa/c$ refers to all the ions from *one mole*, however much solvent they are spread through, and that **increases** on dilution because more of the electrolyte dissociates. So the two quantities move in opposite directions, and the question is testing whether you know which is which.

NCERT · Ch 2, p. 44

1(c)(iii) d- and f-Block

[1]

Transition metals form complex compounds.

Answer · Because their ions are **small and highly charged**, so they have a high charge-to-size ratio and polarise ligands strongly; and because they have **vacant d orbitals of the right energy** to accept lone pairs from ligands.

Why · Both conditions are needed. A small, highly charged ion attracts electron pairs; the vacant $(n-1)d$, ns and np orbitals lie close enough in energy to hybridise and actually accommodate them. Alkali metal ions are large and singly charged with no accessible d orbitals, which is why they form almost no complexes.

NCERT · Ch 4, p. 97

Interhalogen compounds are more reactive than their constituent elements.

Answer · The **X-X' bond in an interhalogen is weaker** than the X-X bond in either halogen (fluorine excepted), because the two atoms differ in size and electronegativity, so the bond is polar and the overlap is poorer. A weaker bond breaks more easily, so the compound is more reactive.

Why · Reactivity here means how readily the molecule dissociates to give halogen atoms or ions. The polarity also makes the larger, less electronegative atom a ready electrophilic centre.

NCERT · — (not in the NCERT Class XII book) *The p-Block Elements chapter was removed from the NCERT Class XII book by the 2023 rationalisation.*

Zinc sulphide is precipitated from an ammoniacal solution in the presence of hydrogen sulphide.

Answer · Ammonia removes H^+ from the solution, which drives the weak-acid equilibrium $H_2S \rightleftharpoons 2H^+ + S^{2-}$ to the **right** and raises $[S^{2-}]$ sharply. The ionic product $[Zn^{2+}][S^{2-}]$ then exceeds the K_{sp} of ZnS, so ZnS precipitates.

Why · In *acidic* solution the opposite happens: the added H^+ suppresses the ionisation of H_2S by the common ion effect, $[S^{2-}]$ stays tiny, and ZnS — which has a relatively large K_{sp} — does not precipitate. CuS, with a far smaller K_{sp} , precipitates even then. That difference is the whole basis of the Group II / Group IV separation in qualitative analysis.

NCERT · — (not in the NCERT Class XII book) *Ionic equilibrium is a Class XI topic and is not in the Class XII book.*

Question 1 (d) · Choose the correct answer

5

The number of unpaired electrons in the complex ion $[CoF_6]^{3-}$ will be: (a) 2 (b) 3 (c) 4 (d) 0

Answer · (c) 4

Why · Co^{3+} is d^6 . F^- is a **weak-field** ligand, so $\Delta_o < P$ and the electrons spread out before pairing: $t_{2g}^4 e_g^2$, which leaves **four** unpaired. The complex is high spin, outer orbital (sp^3d^2) and paramagnetic. With a strong-field ligand such as NH_3 the same d^6 ion would be t_{2g}^6 and diamagnetic.

NCERT · Ch 5, p. 131

The fibre obtained by condensation of hexamethylenediamine and adipic acid is: (a) dacron (b) nylon 66 (c) rayon (d) teflon

Answer · (b) nylon 66

Why · Both monomers carry **six** carbon atoms, which is where the name comes from. The $-NH_2$ and $-COOH$ groups condense with loss of water to give a chain of **amide** linkages — a polyamide. Dacron is a polyester (ethylene glycol + terephthalic acid), rayon is regenerated cellulose and teflon is an addition polymer of tetrafluoroethene.

NCERT · — (not in the NCERT Class XII book) *Polymers was removed from the NCERT Class XII book by the 2023 rationalisation.*

1(d)(iii) Metallurgy

[1]

In the metallurgy of copper, matte is considered to be a mixture of: (a) $\text{FeS} + \text{ZnS}$ (b) $\text{CuS} + \text{FeS}$ (c) $\text{CuFeS}_2 + \text{FeS}$ (d) $\text{Cu}_2\text{S} + \text{FeS}$

Answer · (d) $\text{Cu}_2\text{S} + \text{FeS}$

Why · Roasting copper pyrites converts it to the **cuprous** sulphide, Cu_2S , not the cupric CuS — copper(I) is the stable state in these sulphide melts. The molten mixture of Cu_2S with residual FeS is what is called *matte*; the iron is then removed as a silicate slag in the Bessemer converter.

NCERT · — (not in the NCERT Class XII book) *General Principles and Processes of Isolation of Elements was removed from the NCERT Class XII book by the 2023 rationalisation.*

1(d)(iv) Organic chemistry

[1]

Wöhler's synthesis is used for the preparation of: (a) glycine (b) amino acids (c) urea (d) protein

Answer · (c) urea

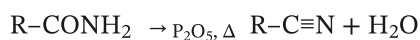
Why · Wöhler heated ammonium cyanate, an inorganic salt, and got urea — $\text{NH}_4\text{CNO} \rightarrow \text{NH}_2\text{CONH}_2$. It is famous for demolishing the vital-force theory, which held that organic compounds could only be made by living things.

NCERT · — (not in the NCERT Class XII book) *Wöhler's synthesis is not named in the Class XII book.*

1(d)(v) Amines / Nitriles

[1]

Dehydration of an amide with P_2O_5 yields: (a) NH_3 (b) RCN (c) RNC (d) RNH_2

**Answer · (b) RCN**

Why · P_2O_5 is a powerful dehydrating agent: it pulls a molecule of water out of R-CONH_2 , leaving the nitrile $\text{R-C}\equiv\text{N}$. Note that the carbon stays bonded to nitrogen through **carbon**, so the product is the cyanide, not the isocyanide. Option (d) would be a *reduction*, and option (c) comes from the carbylamine reaction.

NCERT · Ch 9, p. 263

Part II, Section A · answer any two

Questions 2, 3 and 4, ten marks each. All three are worked.

Q2(a)(i) Solutions

[2]

A solution containing 8 g of compound per 100 mL was found to have an osmotic pressure of 500 cm of Hg at 27 °C. Calculate the molecular mass of the compound.

Formula used · The van't Hoff equation for osmotic pressure, rearranged for molar mass

$$\pi V = nRT = \frac{w}{M} RT \Rightarrow M = \frac{wRT}{\pi V}$$

1. $\pi = \frac{500}{76} = 6.579 \text{ atm}$
2. $V = 0.1 \text{ L}$, $T = 273 + 27 = 300 \text{ K}$, $w = 8 \text{ g}$
3. $M = \frac{8 \times 0.0821 \times 300}{6.579 \times 0.1} = \frac{197.04}{0.6579} = 299.5 \approx 300 \text{ g mol}^{-1}$

Answer · $M \approx 300 \text{ g mol}^{-1}$

Why · Everything is unit conversion. 500 cm Hg must become atmospheres (76 cm Hg = 1 atm), 100 mL must become 0.1 L, and 27 °C must become 300 K, so that $R = 0.0821 \text{ dm}^3 \text{ atm K}^{-1} \text{ mol}^{-1}$ can be used.

NCERT · Ch 1, p. 20

Q2(a)(ii) Solutions

[1]

What happens to the boiling point of water when a small amount of sodium chloride is dissolved in it?

$$\Delta T_b = i K_b m$$

Answer · The boiling point **rises** — elevation of boiling point.

Why · NaCl is non-volatile, so it lowers the vapour pressure of the water. The liquid must then be heated above 100 °C before its vapour pressure again equals atmospheric pressure. NaCl also **dissociates** into two ions, so $i \approx 2$ and the elevation is about twice what the same molality of sugar would give.

NCERT · Ch 1, p. 17

The density of A^+B^- , which has the CsCl (bcc) structure, is 4.4 g/cm^3 . Calculate the edge length of the crystal. [Formula mass = 213]

Formula used · Density of a cubic unit cell; for the CsCl structure there is **one** formula unit per cell, so $Z = 1$

$$d = \frac{Z \times M}{a^3 \times N_A} \Rightarrow a^3 = \frac{Z \times M}{d \times N_A}$$

- $a^3 = \frac{1 \times 213}{4.4 \times 6.022 \times 10^{23}} = 8.04 \times 10^{-23} \text{ cm}^3$
- $a = (8.04 \times 10^{-23})^{1/3} = 4.32 \times 10^{-8} \text{ cm}$

Answer · $a = 4.32 \times 10^{-8} \text{ cm} = 432 \text{ pm}$ (about $4.32 \text{ \AA}$).

Why · The only thing to get right is **Z**. The CsCl arrangement looks body-centred, but the body-centre and the corners are *different* ions, so it is a simple cubic lattice of each ion and the cell contains one Cs^+ ($8 \times \frac{1}{8}$) and one Cl^- (1×1) — **one formula unit**. Taking $Z = 2$ as for a true bcc metal gives 544 pm, which is wrong.

NCERT · — (not in the NCERT Class XII book) The Solid State chapter was removed from the NCERT Class XII book by the 2023 rationalisation.

What is the coordination number of copper in its crystal?

Answer · 12.

Why · Copper crystallises in the **face-centred cubic** (cubic close-packed) lattice. In any close-packed arrangement each sphere touches six in its own layer, three above and three below — twelve in all. That is the maximum possible for equal spheres, and the packing efficiency is 74%.

NCERT · — (not in the NCERT Class XII book) The Solid State chapter was removed from the NCERT Class XII book by the 2023 rationalisation.

An aqueous solution of acetic acid and sodium acetate having $\text{pH} = 4.2$ is diluted to half the initial concentration. What will be the pH of the new solution?

Formula used · The Henderson–Hasselbalch equation for an acidic buffer

$$\text{pH} = \text{pK}_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

Answer · $\text{pH} = 4.2$, unchanged.

Why · Dilution divides the salt and the acid concentrations by the *same* factor, so their **ratio** is unchanged — and only the ratio appears in the Henderson equation. This is precisely the buffer property the question is testing: a buffer resists change in pH on dilution as well as on addition of small amounts of acid or alkali.

NCERT · — (not in the NCERT Class XII book) Buffers are a Class XI topic and are not in the Class XII book.

If one Faraday were 48250 C instead of 96500 C, what will be the charge of an electron?

Formula used · One faraday is the charge on one mole of electrons

$$F = e \times N_A \Rightarrow e = \frac{F}{N_A}$$

$$e = \frac{48250}{6.022 \times 10^{23}} = 8.01 \times 10^{-20} \text{ C}$$

Answer · $e = 8.01 \times 10^{-20} \text{ C}$

Why · The question is really asking whether you know what a faraday is. Avogadro's number is unchanged, so halving F exactly halves the electronic charge — the answer is half of the real $1.602 \times 10^{-19} \text{ C}$.

NCERT · Ch 2, p. 52

Explain the term Ferrimagnetic.

Answer · A **ferrimagnetic** substance is one whose magnetic moments (domains) are aligned in **opposite directions but in unequal numbers**, so the opposing moments do not cancel and the substance has a **small net magnetic moment**. Examples: Fe_3O_4 (magnetite) and the ferrites MFe_2O_4 .

Why · It sits between the two extremes. In a **ferromagnetic** solid all the moments point the same way, giving a large moment; in an **anti-ferromagnetic** solid they are equal and opposite and cancel exactly, giving none. Ferrimagnetics are weakly attracted, and lose their magnetism on heating, becoming paramagnetic.

NCERT · — (not in the NCERT Class XII book) The Solid State chapter was removed from the NCERT Class XII book by the 2023 rationalisation.

The values of λ°_{m} for HCl, NaCl and CH_3COONa are 426.1, 126.5 and $91 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ respectively. Calculate the value of λ°_{m} for acetic acid.

Formula used · Kohlrausch's law of independent migration of ions — combine the three known electrolytes so that the unwanted ions cancel

$$\Lambda^\circ(\text{CH}_3\text{COOH}) = \Lambda^\circ(\text{HCl}) + \Lambda^\circ(\text{CH}_3\text{COONa}) - \Lambda^\circ(\text{NaCl})$$

$$\Lambda^\circ = 426.1 + 91 - 126.5 = 390.6 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

Answer · $\Lambda^\circ_{\text{m}}(\text{CH}_3\text{COOH}) = 390.6 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$

Why · Check the bookkeeping and you can see why the combination works: $(\text{H}^+ + \text{Cl}^-) + (\text{CH}_3\text{COO}^- + \text{Na}^+) - (\text{Na}^+ + \text{Cl}^-)$ leaves exactly $\text{H}^+ + \text{CH}_3\text{COO}^-$. This is the *only* way to get Λ° for a weak electrolyte, because its conductivity cannot be extrapolated to zero concentration directly.

NCERT · Ch 2, p. 49

The temperature coefficient for the hydrolysis of ethyl acetate by sodium hydroxide solution is 1.75. Calculate the energy of activation for the reaction.

Formula used · The temperature coefficient is the ratio of rate constants ten degrees apart, conventionally at 298 K and 308 K; feed it into the two-temperature Arrhenius equation

$$\log \frac{k_{308}}{k_{298}} = \frac{E_a}{2.303 R} \times \frac{T_2 - T_1}{T_1 T_2}$$

1. $\log 1.75 = 0.2430$
2. $E_a = 2.303 \times 8.314 \times 0.2430 \times \frac{298 \times 308}{10}$
3. $E_a = 4.653 \times 9178.4 = 42712 \text{ J mol}^{-1}$

Answer · $E_a \approx 42712 \text{ J mol}^{-1} = 42.7 \text{ kJ mol}^{-1}$

Why · ‘Temperature coefficient’ is simply k_{308}/k_{298} , so the ratio you need is handed to you as 1.75 and no concentrations are involved at all. For most reactions the coefficient is between 2 and 3, so 1.75 is a slightly sluggish reaction and a moderate E_a is the expected answer.

NCERT · Ch 3, p. 82

If 293 K and 303 K are used instead, E_a comes to 41.3 kJ mol^{-1} . Either is acceptable provided the two temperatures are stated.

Deduce the equation to calculate the pH of an acid buffer solution.

Answer · For a weak acid HA with its salt NaA:

$$K_a = [\text{H}^+][\text{A}^-] / [\text{HA}] \Rightarrow [\text{H}^+] = K_a \times [\text{HA}] / [\text{A}^-]$$

Taking $-\log$ of both sides gives the **Henderson–Hasselbalch equation**:

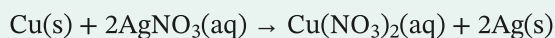
$$\text{pH} = \text{p}K_a + \log ([\text{salt}] / [\text{acid}])$$

Why · Two approximations make it work, and both are justified by the buffer’s composition: the weak acid is so little ionised (further suppressed by the common ion from the salt) that $[\text{HA}]$ equals the acid taken, and the salt is fully ionised so $[\text{A}^-]$ equals the salt taken. It follows immediately that $\text{pH} = \text{p}K_a$ when salt and acid are equal, which is where a buffer works best.

NCERT · — (not in the NCERT Class XII book) Buffers are a Class XI topic and are not in the Class XII book.

AgNO₃ solution cannot be stored in a copper vessel — explain.

Answer · Copper is a stronger reducing agent than silver ($E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34 \text{ V}$ against $E^\circ_{\text{Ag}^+/\text{Ag}} = +0.80 \text{ V}$), so it **displaces silver** from the solution, dissolving the vessel and depositing silver:



Why · $E^\circ_{\text{cell}} = 0.80 - 0.34 = +0.46 \text{ V}$, which is positive, so ΔG° is negative and the reaction is spontaneous. The blue colour of Cu²⁺ appearing in the solution is the visible sign.

NCERT · Ch 2, p. 35

Calculate the molality of a solution prepared by dissolving 3.6 g of glucose in 500 g of water.

Formula used · Molality — moles of solute per **kilogram of solvent**, not per litre of solution

$$m = \frac{\text{moles of solute}}{\text{mass of solvent in kg}} = \frac{w_2 / M_2}{w_1 (\text{kg})}$$

$$m = \frac{3.6 / 180}{0.5} = \frac{0.02}{0.5} = 0.04 \text{ mol kg}^{-1}$$

Answer · $m = 0.04 \text{ mol kg}^{-1}$

Why · Molar mass of glucose C₆H₁₂O₆ = 180 g mol⁻¹, so 3.6 g is 0.02 mol, in 0.5 kg of water.

NCERT · Ch 1, p. 5

Common slip · Molality uses the mass of **solvent**, molarity the volume of **solution**. Molality is preferred in colligative work because mass does not change with temperature.

The standard electrode potentials of Pb/Pb²⁺ and Pt/I⁻/I₂ are -0.126 V and 0.536 V respectively. When a galvanic cell is constructed using 0.1 M concentration of the ions, Pt is found to be the cathode. What is the emf of the cell?

Formula used · Standard cell potential, then the Nernst equation for non-standard concentrations

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}; \quad E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{n} \log Q$$

1. $E^\circ_{\text{cell}} = 0.536 - (-0.126) = 0.662 \text{ V}$
2. $Q = [\text{Pb}^{2+}][\text{I}^-]^2 = 0.1 \times (0.1)^2 = 1 \times 10^{-3}$, so $\log Q = -3$
3. $E_{\text{cell}} = 0.662 - \frac{0.0591}{2} \times (-3) = 0.662 + 0.0887 = 0.751 \text{ V}$

Answer · $E_{\text{cell}} = 0.75 \text{ V}$

Why · Pt being the cathode tells you iodine is reduced and lead is oxidised, so the cell reaction is $\text{Pb} + \text{I}_2 \rightarrow \text{Pb}^{2+} + 2\text{I}^-$, with $n = 2$. Solid Pb and I₂ do not appear in Q.

NCERT · Ch 2, p. 38

What is the Arrhenius equation? How can it be used graphically to calculate the energy of activation for a reaction?

Formula used · The Arrhenius equation, and its linear form

$$k = Ae^{-E_a/RT} \Rightarrow \log k = \log A - \frac{E_a}{2.303 R} \cdot \frac{1}{T}$$

Answer · **The equation** relates the rate constant to temperature: $k = Ae^{-E_a/RT}$, where A is the frequency (Arrhenius) factor, E_a the activation energy and R the gas constant.

Graphically: plot **log k against 1/T**. The linear form above shows this is a straight line of slope $-E_a/2.303 R$ and intercept log A. Measuring the slope therefore gives **$E_a = -2.303 R \times \text{slope}$** .

Why · The exponential term is the *fraction* of molecules with energy at or above E_a . Taking logarithms is what turns an awkward exponential into a straight line, which is why the log k against 1/T plot is the standard experimental method. The slope is negative and E_a positive — the two minus signs cancel.

NCERT · Ch 3, p. 80

The solubility product of BaSO_4 is 1.5×10^{-9} . Find its solubility in 0.1 M BaCl_2 solution.

Formula used · Solubility product with a common ion already present in excess; the Ba^{2+} from the sparingly soluble salt is negligible beside 0.1 M

$$K_{sp} = [\text{Ba}^{2+}][\text{SO}_4^{2-}] \approx (0.1)(s) \Rightarrow s = \frac{K_{sp}}{0.1}$$

$$s = \frac{1.5 \times 10^{-9}}{0.1} = 1.5 \times 10^{-8} \text{ M}$$

Answer · **$s = 1.5 \times 10^{-8} \text{ mol L}^{-1}$**

Why · BaCl_2 is fully ionised and supplies $[\text{Ba}^{2+}] = 0.1 \text{ M}$, which swamps whatever the BaSO_4 itself contributes. Every mole of BaSO_4 that dissolves gives one SO_4^{2-} , so $[\text{SO}_4^{2-}] = s$.

NCERT · — (not in the NCERT Class XII book) Solubility product is a Class XI topic and is not in the Class XII book.

In pure water the solubility would be $\sqrt{K_{sp}} = 3.9 \times 10^{-5} \text{ M}$ — about **2600 times larger**. That is the common ion effect, and it is why a precipitate is washed with a solution of a common ion rather than with water.

(i) How many formula units of sodium chloride are present in a unit cell of sodium chloride? (ii) To what type of cubic system does this cell belong? (iii) What is the geometry of sodium ions around chloride ions?

Formula used · Counting atoms in a cubic cell — a corner is shared by 8 cells, a face centre by 2, an edge centre by 4, a body centre by 1

$$\text{Cl}^-: 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4 \quad \text{Na}^+: 12 \times \frac{1}{4} + 1 = 4$$

Answer · (i) 4 formula units. (ii) **Face-centred cubic** (rock-salt structure) — the Cl^- ions form an fcc lattice with Na^+ in all the octahedral holes. (iii) **Octahedral** — six Na^+ around each Cl^- , so the coordination is 6:6.

Why · Do the counting rather than remembering it. Chloride sits at 8 corners ($\frac{1}{8}$ each) and 6 face centres ($\frac{1}{2}$ each) = 4; sodium sits at 12 edge centres ($\frac{1}{4}$ each) and the body centre (1) = 4. Equal numbers, as the formula NaCl requires, and 4 formula units per cell.

NCERT · — (not in the NCERT Class XII book) The Solid State chapter was removed from the NCERT Class XII book by the 2023 rationalisation.

Part II, Section B · answer any two

Questions 5, 6 and 7, five marks each. All three are worked.

Q5(a) Coordination Compounds

[2]

Write the formulae of the following coordination compounds: **(i)** Potassium dicyanoargentate(I) **(ii)** Hexaaquairon(III) chloride

Answer · (i) $\text{K}[\text{Ag}(\text{CN})_2]$ **(ii)** $[\text{Fe}(\text{H}_2\text{O})_6]\text{Cl}_3$

Why · (i) The *-ate* ending means the complex is the **anion**, so potassium is the counter-ion outside the brackets. Ag is +1 and two CN^- give -2 , so the ion is -1 overall and one K^+ balances it. **(ii)** No *-ate*, so the complex is the **cation**. Six neutral aqua ligands leave the charge as $\text{Fe(III)} = +3$, needing three Cl^- outside.

NCERT · Ch 5, p. 123

Q5(b) Coordination Compounds

[1]

Using C.F.T., determine the number of unpaired electrons for the FeCl_4^- complex.

Answer · 5 unpaired electrons.

Why · Charge balance: $x + 4(-1) = -1$ gives Fe(III), which is **d^5** . The complex is **tetrahedral** (four ligands), and in a tetrahedral field Δ_t is only about $4/9$ of Δ_o — always smaller than the pairing energy. So tetrahedral complexes are **invariably high spin**: $e^2t_2^3$, all five electrons unpaired. Cl^- is a weak-field ligand as well, which points the same way.

NCERT · Ch 5, p. 133

Q5(c) Coordination Compounds

[2]

Explain why tetrahedral nickel complexes are paramagnetic but square planar Ni(II) complexes are diamagnetic.

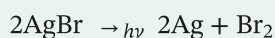
Answer · Ni(II) is **d^8** in both. In a **tetrahedral** complex the splitting Δ_t is small, the ligands are weak-field, and the last two electrons stay **unpaired** in the higher set — two unpaired electrons, so **paramagnetic** (sp^3). In a **square planar** complex a strong-field ligand pushes the $d_{x^2-y^2}$ orbital very high, so all eight electrons pair up in the four lower orbitals leaving it **empty** — no unpaired electrons, so **diamagnetic** (dsp^2).

Why · The geometry is not the cause but the consequence: it is the **strength of the ligand field** that decides both. A strong-field ligand such as CN^- forces pairing, which frees an inner d orbital for dsp^2 and gives square planar; a weak-field ligand such as Cl^- cannot, so the metal falls back on sp^3 and tetrahedral.

NCERT · Ch 5, p. 130

Explain: silver bromide is used in photography.

Answer · AgBr is **light-sensitive**. Light falling on the emulsion reduces the silver ions to metallic silver, forming the invisible *latent image*, which the developer then amplifies:



Unexposed grains are washed away by the fixer (hypo) as the soluble complex $\text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$, leaving a permanent image.

Why · The photo-decomposition works because Ag^+ is easily reduced and the Ag–Br bond has enough covalent character to absorb visible light. AgCl behaves similarly but is less sensitive; AgI is used in the fastest emulsions.

NCERT · Ch 5, p. 137 — mentioned under the uses of coordination compounds

Explain: in a given transition series there is no significant change in the atomic radii of the elements with increase in atomic number.

Answer · Across a transition series, each added proton increases the nuclear charge and pulls the shell in, but each added electron goes into the inner **(*n*–1)*d*** subshell, which **screens** the outermost *ns* electrons from that extra charge. The two effects very nearly cancel, so the effective nuclear charge felt by the outer electrons — and hence the radius — changes very little.

Why · *d* electrons screen poorly compared with *s* and *p* electrons, but they screen well enough for the balance to hold across the middle of the series. The radius does fall slightly at the start and rise again near the end, where electron–electron repulsion in the filling *d* orbitals begins to win.

NCERT · Ch 4, p. 93

Draw the structure of xenon oxyfluoride molecules.

Answer · **XeOF₄** — sp^3d^2 , six electron pairs (5 bond + 1 lone) → **square pyramidal**, with the O at the apex, four F in the plane and the lone pair below.

XeO₂F₂ — sp^3d , five electron pairs (4 bond + 1 lone) → **see-saw** (trigonal bipyramidal with one equatorial position occupied by the lone pair); the two F are axial and the two O equatorial.

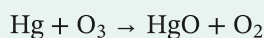
XeO₃F₂ — sp^3d , five bond pairs and no lone pair → **trigonal bipyramidal**, with the two F axial and the three O equatorial.

Why · Count with VSEPR each time: total electron pairs decide the *arrangement*, and the lone pairs then decide the *shape* by taking the roomiest positions — equatorial in a trigonal bipyramid, because an axial lone pair would suffer three 90° repulsions instead of two.

NCERT · — (not in the NCERT Class XII book) The *p-Block Elements* chapter was removed from the NCERT Class XII book by the 2023 rationalisation.

Give balanced equations for: **(i)** ozone and mercury; **(ii)** chlorine passed through hot concentrated NaOH solution; **(iii)** silver nitrate added to a dilute solution of sodium thiosulphate.

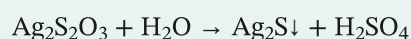
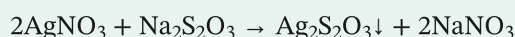
Answer · **(i)** Ozone oxidises mercury, which loses its meniscus and sticks to glass — the *tailing of mercury*:



(ii) With **hot concentrated** alkali chlorine disproportionates to chloride and **chlorate**:



(iii) A white precipitate of silver thiosulphate forms and then darkens through yellow and brown to black Ag_2S :

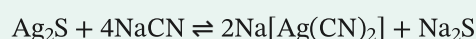


Why · (ii) is the one to be careful with: **cold dilute** NaOH gives chloride and **hypochlorite** (3Cl_2 would be $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaOCl} + \text{H}_2\text{O}$), whereas hot concentrated alkali carries the oxidation further to chlorate. The conditions decide the product, and the question names them deliberately. (iii) needs the thiosulphate to be **dilute**; in excess thiosulphate the soluble complex forms instead, which is how hypo fixes a photograph.

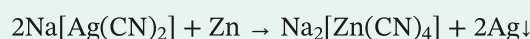
NCERT · — (not in the NCERT Class XII book) *The p-Block Elements chapter was removed from the NCERT Class XII book by the 2023 rationalisation.*

How is silver extracted from its sulphide ore by the cyanide process? Give only the reactions.

Answer · **Leaching** — the ore is treated with dilute NaCN in the presence of air, which dissolves the silver as a soluble dicyanoargentate(I) complex:



Displacement — the silver is then precipitated with the more electropositive zinc:



Why · The first equilibrium is unfavourable on its own; air is bubbled through to oxidise the Na_2S away, which pulls the equilibrium to the right. The second step works because zinc is above silver in the electrochemical series. It is the same Mac Arthur–Forrest process used for gold.

NCERT · — (not in the NCERT Class XII book) *General Principles and Processes of Isolation of Elements was removed from the NCERT Class XII book by the 2023 rationalisation.*

Part II, Section C · answer any two

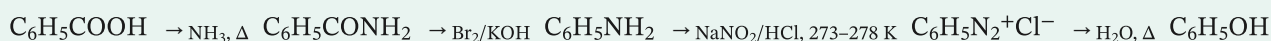
Questions 8, 9 and 10, ten marks each — all organic. All three are worked.

Q8(a) Organic conversions

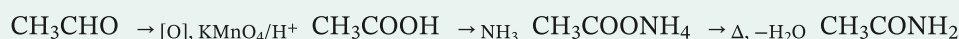
[6]

How can the following conversions be brought about? **(i)** Benzoic acid to phenol **(ii)** Acetaldehyde to acetamide **(iii)** Aniline to fluorobenzene **(iv)** Propionic acid to acetic acid

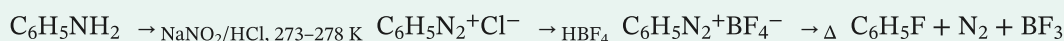
Answer · (i) Benzoic acid → phenol (four steps, through the diazonium salt):



(ii) Acetaldehyde → acetamide:



(iii) Aniline → fluorobenzene (Balz–Schiemann reaction):



(iv) Propionic acid → acetic acid (a step-down, losing one carbon):



Why · Count the carbons first — it tells you which tool you need. (i) and (iv) both lose one carbon, and the standard way to do that is the **Hofmann bromamide degradation**, which converts an amide to an amine with one carbon fewer. (iii) cannot be done by direct fluorination, which is far too violent; the diazonium fluoroborate route is the only practical way to put F on a ring.

NCERT · Ch 9, p. 264

Q8(b) Amines / Nitriles

[2]

Identify the products A, B, C and D: $\text{CH}_3\text{CH}_2\text{Br} \rightarrow \text{KCN}$ A $\rightarrow \text{H}_2\text{O}/\text{H}^+$ B $\rightarrow \text{PCl}_5$ C $\rightarrow \text{NH}_3$ D

Answer · A = $\text{CH}_3\text{CH}_2\text{CN}$, propanenitrile · **B** = $\text{CH}_3\text{CH}_2\text{COOH}$, propanoic acid · **C** = $\text{CH}_3\text{CH}_2\text{COCl}$, propanoyl chloride · **D** = $\text{CH}_3\text{CH}_2\text{CONH}_2$, propanamide.

Why · KCN is ionic, so the free CN^- attacks through **carbon** and the chain grows by one — two carbons in, three carbons out. Acid hydrolysis then takes the nitrile to the carboxylic acid, PCl_5 replaces $-\text{OH}$ by $-\text{Cl}$, and ammonia replaces $-\text{Cl}$ by $-\text{NH}_2$.

NCERT · Ch 6, p. 170

Common slip · Had the reagent been **AgCN**, A would have been the *isocyanide* $\text{CH}_3\text{CH}_2\text{NC}$ and the whole chain would be different — the same distinction the 2024–25 paper tests in Q1(B)(vi).

Complete the given reactions: (i) $\text{CH}_3\text{COCH}_3 + \text{I}_2 + \text{KOH} \rightarrow$ (ii) $\text{C}_6\text{H}_5\text{NH}_2 + \text{CH}_3\text{COCl} \xrightarrow{\text{pyridine}}$

Answer · (i) The iodoform reaction — a yellow precipitate of CHI_3 :



(ii) Acetylation — the product is acetanilide:



Why · (i) Acetone has the $\text{CH}_3\text{CO}-$ grouping, which is what the iodoform test requires. **(ii)** Pyridine is there to mop up the HCl produced; without it the HCl would protonate the aniline to the anilinium salt, whose nitrogen is no longer nucleophilic, and the reaction would stall.

NCERT · Ch 9, p. 272

Give balanced equations for: (i) Kolbe–Schmidt reaction (ii) Hell–Volhard–Zelinsky reaction (iii) Clemmensen's reduction

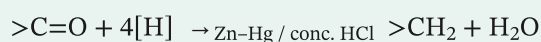
Answer · (i) Kolbe–Schmidt — note it starts from the **phenoxide**:



(ii) Hell–Volhard–Zelinsky — halogenation at the α -carbon:



(iii) Clemmensen's reduction — carbonyl all the way to CH_2 :



Why · HVZ needs an α -hydrogen, so it fails on HCOOH and on benzoic acid. Clemmensen works in *acid*; the Wolff–Kishner reduction ($\text{NH}_2\text{NH}_2/\text{KOH}$) does the same job in *base*, and you choose between them by what else is in the molecule.

NCERT · Ch 8, p. 253

An organic compound [A] with molecular formula C_3H_6O , which reduces Tollens' reagent, gives [B] with hydroxylamine. [B] on reduction with $LiAlH_4$ gives [C]. [C] on warming with chloroform and potassium hydroxide gives [D] with an unpleasant smell. [D] on reduction with Na/C_2H_5OH gives [E], which gives an insoluble compound with benzenesulphonyl chloride and $NaOH$. Give the sequence of reactions and suggest structures of [A] to [E].

1. $CH_3CH_2CHO + NH_2OH \rightarrow CH_3CH_2CH=NOH$ (B) + H_2O
2. $B + 4[H] \xrightarrow{LiAlH_4} CH_3CH_2CH_2NH_2$ (C) + H_2O
3. $C + CHCl_3 + 3KOH \rightarrow CH_3CH_2CH_2NC$ (D) + $3KCl + 3H_2O$ — carbylamine
4. $D + 4[H] \xrightarrow{Na/C_2H_5OH} CH_3CH_2CH_2NHCH_3$ (E)

Answer · A = CH_3CH_2CHO , propanal · B = $CH_3CH_2CH=N-OH$, propanal oxime · C = $CH_3CH_2CH_2NH_2$, propan-1-amine · D = $CH_3CH_2CH_2NC$, propyl isocyanide · E = $CH_3CH_2CH_2NHCH_3$, N-methylpropan-1-amine.

Why · Each clue pins one structure. **Reduces Tollens'** → A is an *aldehyde*, and C_3H_6O leaves only propanal (the isomer acetone is a ketone and would not). Hydroxylamine gives an **oxime**; $LiAlH_4$ reduces $C=N$ to $CH-NH_2$, giving a **primary** amine. **Chloroform and KOH with an unpleasant smell** is the **carbylamine test**, which only primary amines give — confirming C. Reducing an isocyanide gives a **secondary** amine, and the last clue closes the loop: in the **Hinsberg test** a 2° amine gives a sulphonamide with no N-H left, so it cannot dissolve in alkali — the *insoluble* compound.

NCERT · Ch 9, p. 271

Give one chemical test to distinguish between: (i) Formic acid and oxalic acid (ii) Glucose and fructose

Answer · (i) Add **calcium chloride solution** (to the neutralised acid): **oxalic acid** gives a **white precipitate** of calcium oxalate, which is insoluble in acetic acid; **formic acid** gives none. Alternatively, warm with **Tollens' reagent**: formic acid gives a **silver mirror**, oxalic acid does not.
(ii) **Bromine water**: **glucose**, an aldose, is oxidised to gluconic acid and **decolourises** it; **fructose**, a ketose, does **not**. (Seliwanoff's test works the other way — fructose gives a red colour rapidly.)

Why · (i) Formic acid, $HCOOH$, is the only carboxylic acid that also contains a $-CHO$ group, so it alone reduces Tollens'. Oxalic acid is distinguished instead by the insolubility of its calcium salt. (ii) Bromine water is a **mild** oxidant: it oxidises an aldehyde group but not a ketone. This is exactly why bromine water — and not Tollens' or Fehling's — distinguishes the two sugars: in alkaline Tollens' or Fehling's, fructose **isomerises** to glucose and gives a positive test too.

NCERT · Ch 10, p. 283

Common slip · Naming Tollens' or Fehling's for (ii) is the standard error. Both are alkaline, and alkali converts fructose to glucose, so both sugars respond.

Write balanced reactions for: **(i)** acetamide heated with phosphorus pentoxide followed by hydrolysis; **(ii)** benzoic acid treated with a mixture of conc. H_2SO_4 and conc. HNO_3 ; **(iii)** ethyl magnesium chloride treated with formaldehyde and the product hydrolysed in the presence of an acid; **(iv)** diethyl ether reacts with excess of HI.

Answer · **(i)** Dehydration to the nitrile, then hydrolysis back to the acid:



(ii) $-\text{COOH}$ is deactivating and **meta-directing**:



(iii) Formaldehyde with any Grignard gives a **primary** alcohol:



(iv) With **excess** HI both halves become the iodide:



Why · **(ii)** is the one that tests understanding rather than recall: $-\text{COOH}$ withdraws electrons, so the ring is deactivated and the electrophile is directed to the *meta* position — not *ortho/para*. **(iv)** turns on the word **excess**: one equivalent of HI gives ethanol and ethyl iodide, and it is the second equivalent that converts the ethanol to the iodide as well.

NCERT · Ch 7, p. 219

On electrolysis in acidic solution amino acids migrate towards the cathode, while in alkaline solution they migrate towards the anode. Explain.

Answer · An amino acid exists as the **zwitter ion**, $^+\text{H}_3\text{N}-\text{CHR}-\text{COO}^-$, which is overall neutral. **In acid** the excess H^+ protonates the $-\text{COO}^-$, leaving a net **positive** cation, which moves to the **cathode**. **In alkali** the OH^- removes a proton from $-\text{NH}_3^+$, leaving a net **negative** anion, which moves to the **anode**.

Why · The amino acid is **amphoteric** — it has an acidic and a basic group on the same molecule, so it responds to whichever is offered. At the intermediate pH where the two cancel exactly, the **isoelectric point**, the molecule is electrically neutral and does not migrate at all; that pH is also where its solubility is at a minimum, which is how proteins are separated by electrophoresis.

NCERT · Ch 10, p. 292

Deficiency of which vitamin will cause: **(i)** night blindness **(ii)** rickets?

Answer · **(i) Vitamin A** **(ii) Vitamin D**

Why · Vitamin A (retinol) is the precursor of rhodopsin, the pigment in the retina that responds to dim light. Vitamin D governs the absorption of calcium from the gut, so its deficiency gives soft, deformed bone — rickets in children, osteomalacia in adults. Both are **fat-soluble**, which is why they can be stored and why excess is harmful.

NCERT · Ch 10, p. 296

Name the type of polymerisation and the monomers in each of the following polymers: (i) Dacron (ii) Nylon-66

Answer · (i) Dacron (Terylene) — **condensation** (step-growth) polymerisation; monomers **ethylene glycol** and **terephthalic acid**. It is a **polyester**.

(ii) Nylon-66 — **condensation** (step-growth) polymerisation; monomers **hexamethylenediamine** and **adipic acid**. It is a **polyamide**.

Why · Both are condensation polymers because each linkage forms with the **loss of a small molecule** (water). The difference is only in which groups meet: -OH with -COOH gives an **ester** link, -NH_2 with -COOH gives an **amide** link. Contrast addition polymers such as polythene, where nothing is lost.

NCERT · — (not in the NCERT Class XII book) Polymers was removed from the NCERT Class XII book by the 2023 rationalisation.